Advanced Calculus I Non-hand-in Assignment 1

This homework comes from textbook and will **NOT** be collected (這份作業完成後不需要繳交、評分。). But for practice,or quizzes or exams, please make sure that you have idea to solve the following questions by yourself. (這些題目是很重要的練習，小考、期中、期末考會有類似的題目。練習這些題目的過程中所逐漸形成我們對於所學到的知識的"看法"，會在考試中很有幫助。所以，請一定要了解如何解題與解題背後的想法是什麼。)

Please prove the following propositions are true. (請證明下列命題爲眞。)

**Section 1.2.**
Example 1.2 (on page 23).
1.6 Theorem (on page 25).

# 第 1 週｜自主練習・課本題目補充

William R. Wade | An Introduction to Analysis | Fourth Edition, Global Edition

依教師 Non-hand-in Assignment 指定順序整理；以下為題目／命題敘述，不含解答。

## Example 1.2

Wade p.23

**1.2** ***EXAMPLE.***

If $a \in \mathbf{R}$, prove that

$$a \neq 0 \quad \text{implies} \quad a^2 > 0. \tag{5}$$

In particular, $-1 < 0 < 1$.

## Theorem 1.6

Wade p.25

**1.6 Theorem.** **[FUNDAMENTAL THEOREM OF ABSOLUTE VALUES].**
*Let $a \in \mathbf{R}$ and $M \geq 0$. Then $|a| \leq M$ if and only if $-M \leq a \leq M$.*